Handwritten digit recognition based on LeNet and its visualization analysis

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Abstract

Deep learning neural networks are generally opaque, meaning that although they can make useful and skillful predictions, it is not clear how or why a given prediction was made.

Convolutional neural networks, have internal structures that are designed to operate upon two-dimensional image data, and as such preserve the spatial relationships for what was learned by the model[5]. Specifically, the two-dimensional filters learned by the model can be inspected and visualized to discover the types of features that the model will detect, and the activation maps output by convolutional layers can be inspected to understand exactly what features were detected for a given input image.

Our work involves utilizing LeNet to recognize handwritten digits in the MNIST dataset. We analyze the recognition results of LeNet and attempt to explain them by visualizing various components of the LeNet network. We visualize the model architecture, filters, and feature maps of LeNet separately, aiming to analyze their respective roles in the neural network.

Keywords: LeNet, visualization

1 introdcution

1.1 MNIST

The MNIST database is a large database of handwritten digits that is commonly used for training various image processing systems[1]. The training dataset was taken from American Census Bureau employees, while the testing dataset was taken from American high school students. The MNIST database contains 60,000 training images and 10,000 testing images. Half of the training set and half of the test set were taken from NIST's training dataset.Here is an overview of MNIST in figure below and it's information:

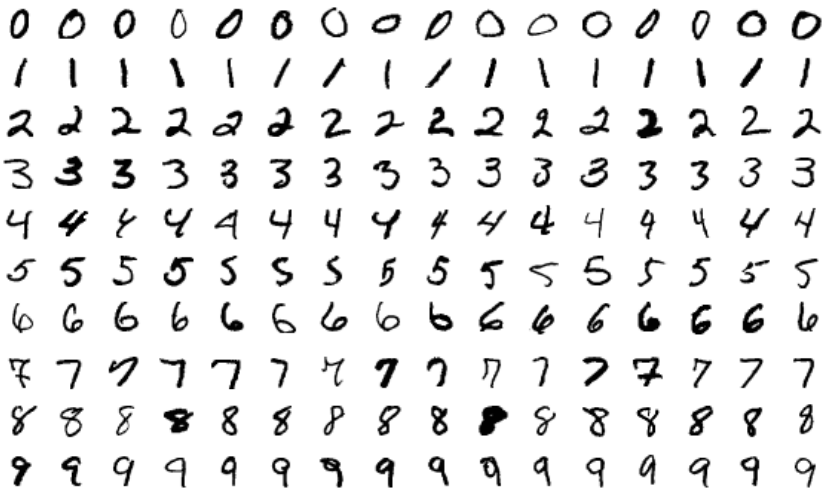


Figure 1: example of MNIST

1.2 LeNet

LeNet-5 is a convolutional neural network(CNN) structure proposed by LeCun et al[3]. in 1998. It was one of the earliest CNNs and promoted the development of deep learning.

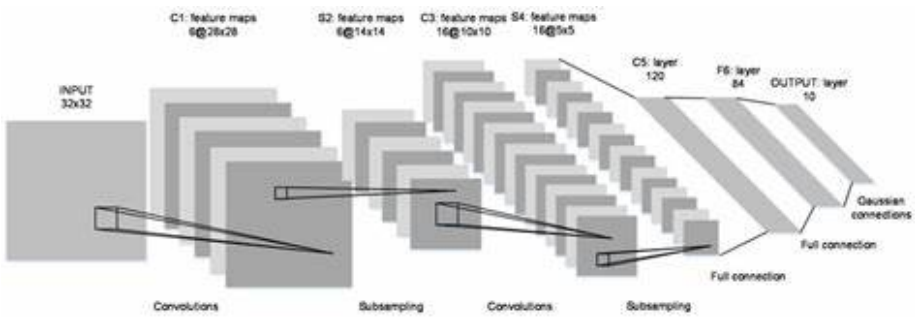


Figure 2: LeNet structure

As shown in the figure, LeNet-5 consists of seven layers(2 convolution layers, 2 pooling layers, 3 fully connected layers). In addition to pooling layers, every other layer can train parameters. In the figure, Cx represents convolution layer, Sx represents sub-sampling layer, Fx represents complete connection layer, and x represents layer index.

Layer C1 is a convolution layer with six convolution kernels of 5x5 and the size of feature mapping is 28x28, which can prevent the information of the input image from falling out of the boundary of convolution kernel.

Layer S2 is the pooling layer that outputs 6 feature graphs of size 14x14.

Each cell in each feature map is connected to 2x2 neighborhoods in the corresponding feature map in C1.

Layer C3 is a convolution layer with 16 5x5 convolution kernels. The input of the first six C3 feature maps is each continuous subset of the three feature maps in S2, the input of the next six feature maps comes from the input of the four continuous subsets, and the input of the next three feature maps comes from the four discontinuous subsets. Finally, the input for the last feature graph comes from all feature graphs of S2.

Layer S4 is similar to S2, with size of 2x2 and output of 16 5x5 feature graphs.

Layer FL5 is a flatten layer which converts 4x4x16 feature graphs into a vector of length 256 as the input of the rest fully connected part.

F6 layer is fully connected to FL5, and 120 feature graphs are output. The same like layer F7, F8 with respectively output sizes 84 and 10.

2 Results and Analysis

2.1 Classification Results

In our experiments, we evaluate the network performance from four aspects: accuracy, Cross-entropy loss[4] and the number of parameters.

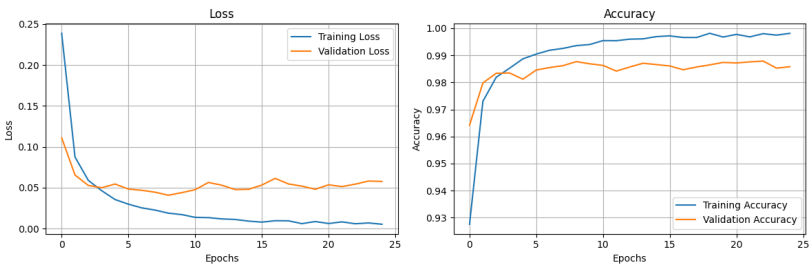


Figure 3: Error and accuracy curves over training epochs

As shown in the figure, after 25 epochs of training, the model's accuracy and cross-entropy loss on both the training and test sets become nearly equal, indicating the onset of overfitting. Beyond 10 epochs, there is little change in the model's accuracy and loss on both sets. After 25 epochs of training, LeNet achieves an accuracy of 98.58% on the test set with a loss of 0.057.

From the confusion matrix[2], we observe that LeNet exhibits good classification performance for most digits. The most frequently misclassified digits are 6 and 4. Interestingly, these two classes are indeed challenging to distinguish even for humans, as evident from the sample images we have seen previously.

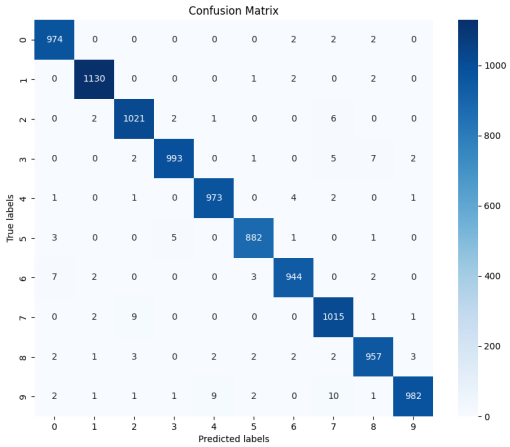


Figure 4: Confusion Matrix

We used the tkinter library and the Keras framework to implement testing of our own handwritten digits. The test results are as follows

Figure 5: gui

2.2 Model Architecture Visualization

We used the'kerasutils.plot_model()'function to visualize the model architecture, as shown in the figure. The number of parameters in the LeNet model is calculated as follows:

$$(25+1)\times 6+(25\times 6+1)\times 16+(48000+120)+(100800+84)+(840+10)=62006$$

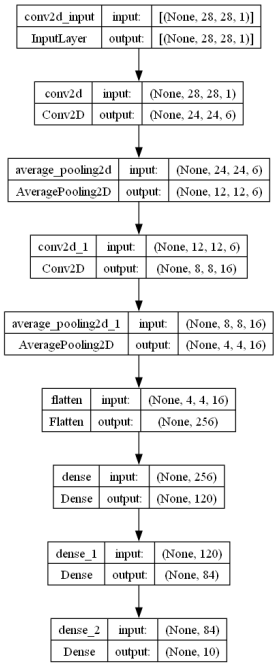


Figure 6: LeNet Structure

2.3 Filter and Feature Map Visualization

The convolutional layer comprises two layers, denoted as C1 and C3, with their weights linearly normalized and mapped to the range of $0 \sim 255$ for visualization, as shown in the figure. Darker shades indicate smaller weights, while lighter shades indicate larger weights.

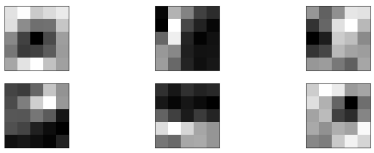


Figure 7: The six convolutional kernels of the first convolutional layer

The C1 layer consists of 6 convolutional kernels, extracting lower-level features. We observe that kernel a has relatively larger weights around its edges. For the input image of digit 0 depicted in the illustration, its corresponding extracted feature map a' primarily focuses on the edge regions of the image. Kernel b exhibits lighter colors on its left half and darker colors on its right half, suggesting that it extracts vertical edge features. Accordingly, its resulting feature map b' displays distinct vertical boundary lines. Kernel c displays

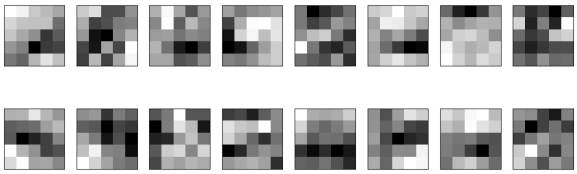


Figure 8: The sixteen convolutional kernels of the second convolutional layer

darker colors on its upper half and lighter colors on its lower half, indicating it extracts horizontal edge features. In the corresponding feature map c', noticeable horizontal edge details are extracted in the lower half.

Since the C3 layer extracts high-dimensional features, it may appear abstract to the human eye. Therefore, let's compare the test results of digits with similar local structures to gain insights.

2.3.1 Analysis of the first feature map(row1, col3)

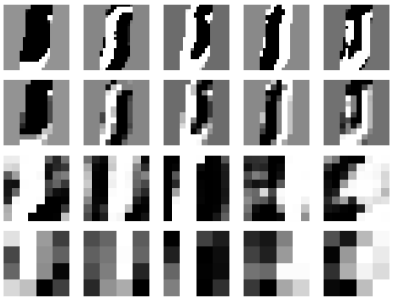


Figure 9: featuremaps of 1

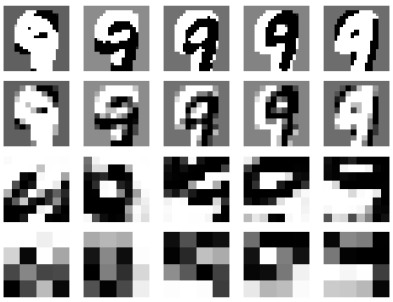


Figure 10: featuremaps of 9

Comparison of feature maps of digits 1 and 9 reveals prominent vertical

details, similarly observed in digits 4 and 7. The horizontal parts are completely suppressed while the vertical parts are extracted. Therefore, we conclude that the first feature map extracts the vertical local structures of the images.

2.3.2 Analysis of the second feature map(row2, col3)

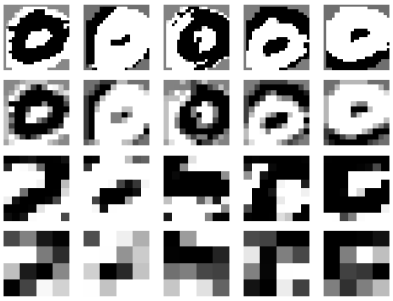


Figure 11: featuremaps of 1

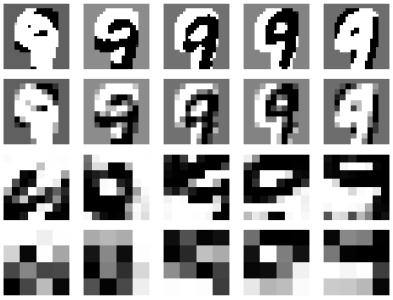


Figure 12: featuremaps of 9

Comparison of the feature maps of digits 9 and 0 reveals prominent circular structures, as both digits contain circular components. Therefore, we can observe that the second feature map extracts distinct circular structures. In contrast, when comparing the second feature maps of digits 2 and 3, we find that they capture regions with significant curvature but lack the standard circular structure seen in digits 9 and 0. Hence, we infer that this feature map extracts local structures with significant curvature.

2.4 The impact of different activation functions and receptive fields on visualization effects

2.4.1 The Influence of Different Activation Functions on Visualization Effects

We conducted separate tests using tanh, relu, and sigmoid as activation functions. The visualization results obtained are as follows:

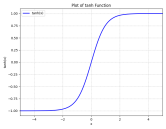
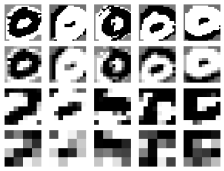
[a]  [b] [c] 

Figure 13: tanh

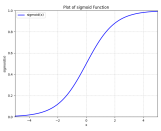
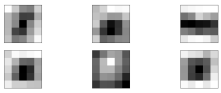
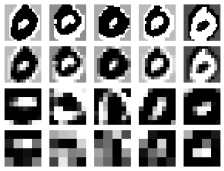
[a]  [b]  [c] 

Figure 14: sigmoid

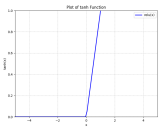
[a]  [b] [c]

Figure 15: reLu

We can observe that although different activation functions lead LeNet to learn different filters and thus produce diverse feature maps, they share commonality in extracting similar types of information. Specifically, we can identify kernels that extract vertical edge information and kernels resembling Laplacian operators, which are well-known for detecting edges and sharp transitions in images.

2.4.2 The Influence of Different Receptive Field Sizes on Visualization Effects

By adjusting the size of the receptive field, we obtain the modified filters and feature maps as follows:

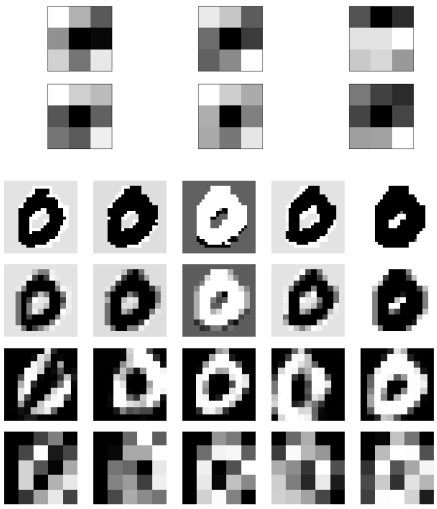


Figure 16: Receptive field size of 3x3

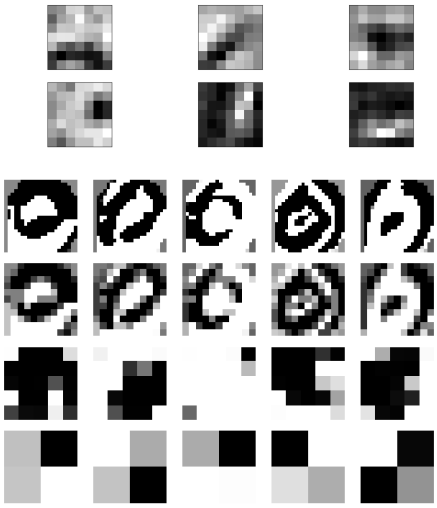


Figure 17: Receptive field size of 3x3

We can observe that after reducing the size of the convolutional kernel, the similarity between different feature maps significantly increases. Conversely, when enlarging the convolutional kernel, some kernels display new information: for example, certain kernels show a preference for direction (as seen in the second kernel) or circles (as observed in the fourth kernel), which is also reflected in the feature maps.

3 Conclusion

In summary, we have successfully implemented handwritten digit classification tasks using LeNet. We visualized the model architecture, filters, and feature maps and provided explanations for their visualization effects. Additionally, we compared the visualization results obtained using different activation functions and convolutional kernel sizes. However, we acknowledge certain limitations. Due to the small size and grayscale nature of the MNIST dataset images, the visualization results of the filters may not accurately reflect which specific directional and color features are being extracted. Furthermore, our analysis of higher-level convolutional kernels and feature maps may still be limited and requires further research and in-depth analysis.

4 Research methods

4.1 Collecting data

This project uses Microsoft Kinect Studio for recording IR 11 Bit data. The newer version SDK allows to capture recorded data in a processed event file format (XEF) allows to record processed IR data as well as depth and skeleton data. I f data is recorded in XRF format then it could be converted by using the KSConvert in a command line prompt available in the Kinect files folders. To get a satisfactory result out of the recording we will capture recording by allowing each members of the target audience individually when they are interacting with the prototyping of our application. The different tilt angels of the sensor and having various outfits specially upper parts of wearing cloths will be considered during our recording settings.

5 Task Distribution

Taks distributions are as follow:

Building LeNet and recognition: Huang YiKe

Visualization of the results, filters and featuremaps: Hu Junhao

GUI: Huang Yike

PPT: HuJunhao

Report: HuJunhao, Huang Yike

References

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